

Tema 1: Transformada de Fourier

(1) Obtener la transformada de Fourier de los siguientes ejercicios aplicando propiedades

1.

$$x(t) = u(t - 2) - u(t - 6)$$

$$X(\omega) = \frac{2\operatorname{sen}(2\omega)}{\omega} e^{-j4\omega}$$

2.

$$x(t) = e^{-6|t|}$$

$$X(\omega) = \frac{12}{36 + \omega^2}$$

3.

$$x(t) = \frac{10}{5 + jt}$$

$$X(\omega) = 20\pi u(-\omega) e^{5\omega}$$

4.

$$x(t) = \frac{1}{9 + t^2}$$

$$X(\omega) = \frac{1}{3} \pi e^{-3|\omega|}$$

5.

$$x(t) = e^{4jt} u(t) e^{-8t}$$

$$X(\omega) = \frac{1}{8 + (\omega - 4)j}$$

6.

$$x(t) = e^{3jt}(u(t+2) - u(t-2))$$

$$X(\omega) = \frac{2sen(2(\omega-3))}{\omega-3}$$

7.

$$x(t) = 5e^{-7|t+2|}$$

$$X(\omega) = \frac{70}{49 + \omega^2} e^{j2\omega}$$

8.

$$x(t) = t^2 e^{-t^2}$$

$$X(\omega) = \sqrt{\pi} \frac{2 - \omega^2}{4} e^{\frac{-\omega^2}{4}}$$

9.

$$x(t) = (t-2)u(t-2)e^{-6t}$$

$$X(\omega) = \frac{e^{-2(6+j\omega)}}{(6+j\omega)^2}$$

10.

$$x(t) = t(u(t+3) - u(t-3))$$

$$H(\omega) = \frac{1}{j\omega} \left[-6\cos(3\omega) + \frac{2sen(3\omega)}{\omega} \right] + 4\pi\delta(\omega)$$

(2) Obtener la $y(t)$ de las siguientes ecuaciones diferenciales.

1.

$$y'(t) - 6y(t) = x(t), \text{ para } x(t) = u(t)e^{-6t}$$

$$y(t) = \frac{1}{12} e^{-6|t|}$$

2.

$$4y - y''(t) = x(t) , \text{ para } x(t) = e^{-10|t|}$$

$$y(t) = \frac{5}{96}e^{2t}\mu(t) - \frac{5}{96}e^{-2t}\mu(t) - \frac{1}{96}e^{-10t}\mu(t) + \frac{1}{96}e^{10t}\mu(t)$$

3.

$$4y + 2y' - y''(t) = x(t) , \text{ para } x(t) = e^{-3|t|}$$

$$y(t) = -2.05e^{3t}\mu(t) - 0.18e^{-3t}\mu(t) + 0.36e^{-1.24t}\mu(t) - 1.87e^{3.23t}\mu(t)$$

4.

$$-y'' + y = x(t) , \text{ para } x(t) = \frac{1}{8}u(t)e^{-2t}$$

$$y(t) = -\frac{1}{48}e^{-t}\mu(t) - \frac{1}{48}e^t\mu(t) - \frac{1}{24}e^{2t}\mu(t)$$

(3) Obtener $y(t)$ de la convolución entre $x(t)$ y $h(t)$.

1.

$$x(t) = 2 + \cos(2t) + 2 \cdot \cos(t/2)$$

$$h(t) = 2 \frac{\sin(t)}{t}$$

$$y(t) = 4\pi \cos(t/2) + 4\pi$$

2.

$$x(t) = \cos(t) + 2 \cdot \cos(t/2)$$

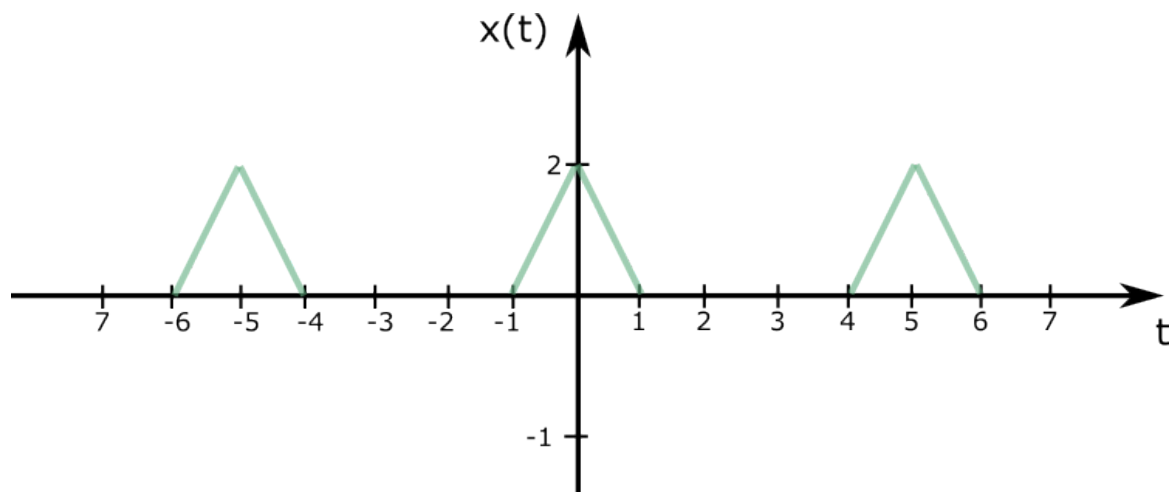
$$h(t) = 2 \frac{\sin(2t)}{t}$$

$$y(t) = 2\pi \cos(t) + 4\pi \cos(t/2)$$

Tema 2: Serie/Transformada de Fourier

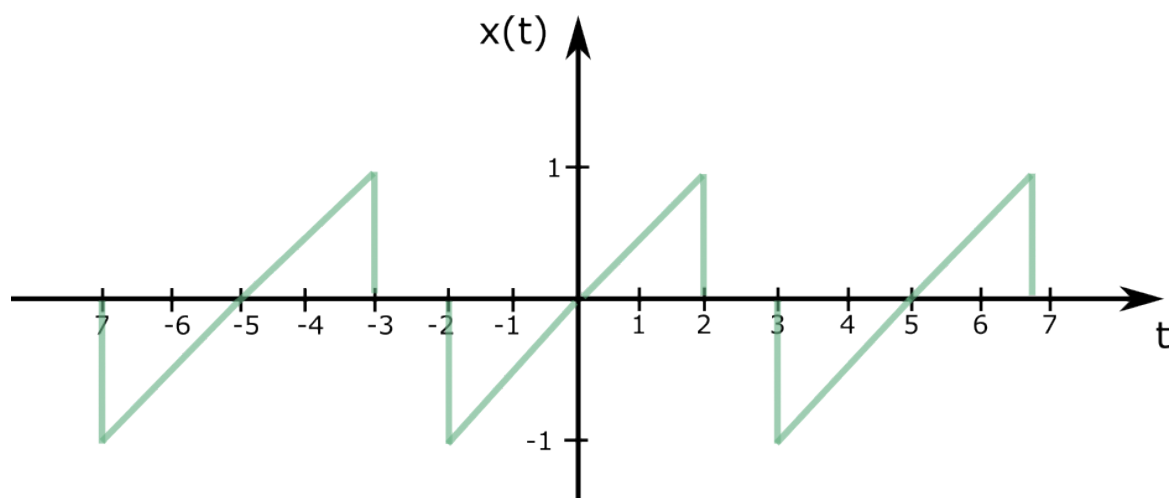
(1) Obtener los coeficientes de la serie de Fourier a través de la transformada de Fourier de los ejercicios de la guía anterior, comparar los resultados y complejidad de la resolución.

1.



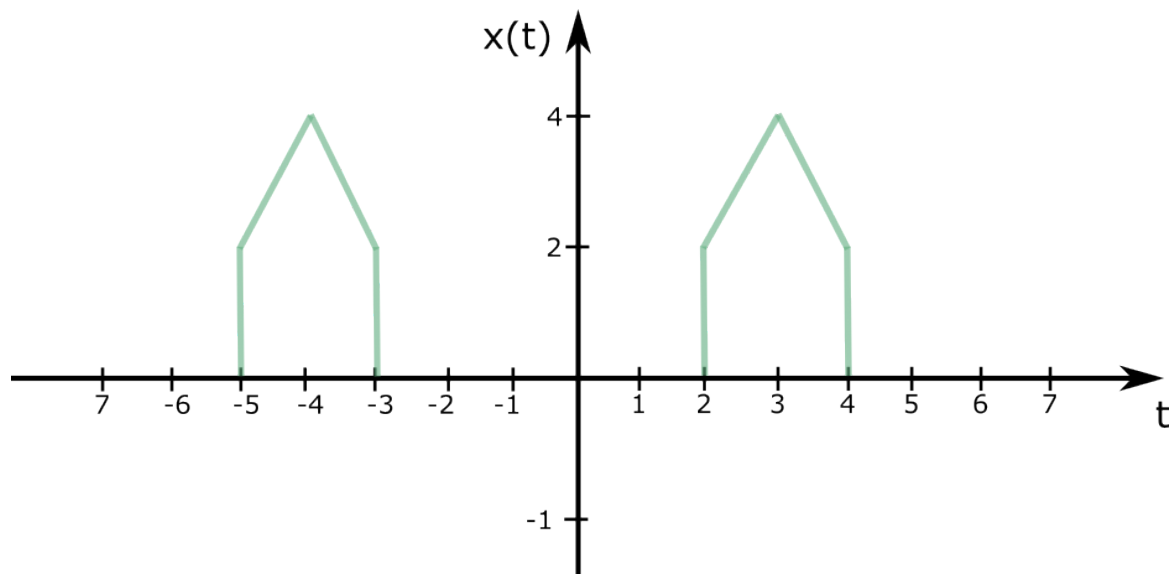
$$a_k = \frac{50 \sin^2(k\pi/5)}{(k\pi)^2}, \quad a_0 = 2, \quad T = 5, \quad \omega_0 = \frac{2\pi}{5}$$

2.



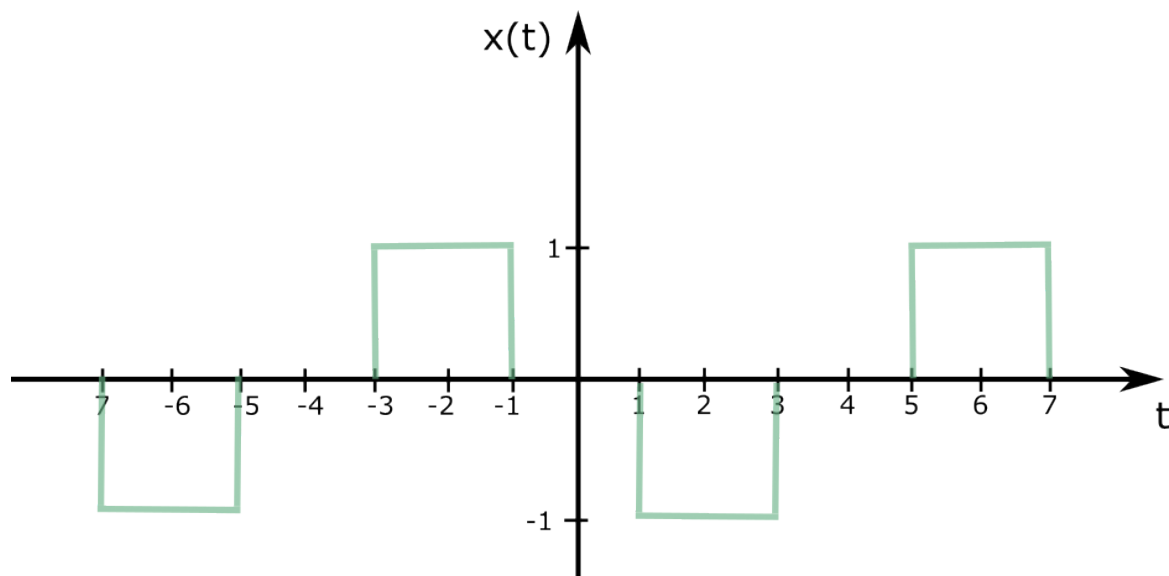
$$a_k = j \frac{1}{6} \left[\frac{3 \cos(k2\pi/3)}{k\pi} - \frac{9 \sin(k2\pi/3)}{(k\pi)^2} \right], \quad a_0 = 0, \quad T = 5, \quad \omega_0 = 2\pi/5$$

3.



$$a_k = e^{-jk6\pi/7} \left[\frac{\text{sen}(k2\pi/7)}{k\pi} + \frac{98\text{sen}^2(k\pi/7)}{(k\pi)^2} \right], \quad a_0 = 8, \quad T = 7, \quad \omega_0 = 2\pi/7$$

4.

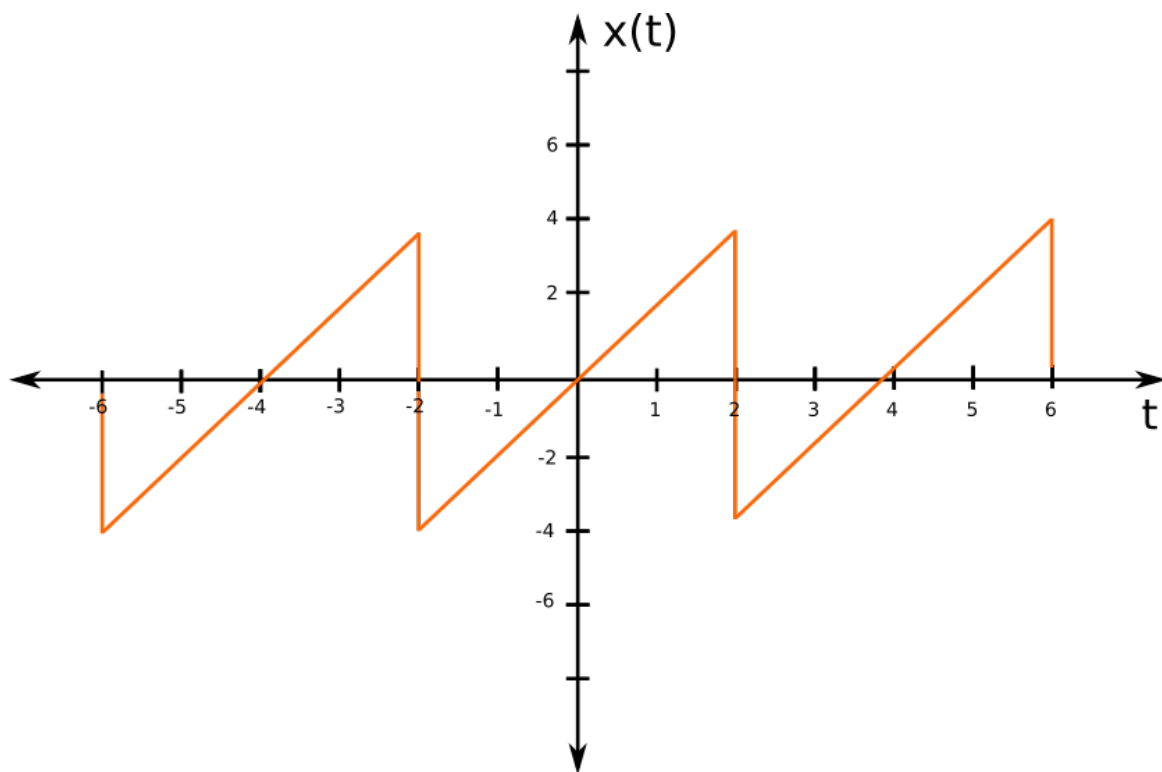


$$a_k = j2 \frac{\text{sen}^2(k\pi/4)}{k\pi}, \quad a_0 = 0, \quad T = 8, \quad \omega_0 = \pi/4$$

Tema 3: Sistemas SLIT y Serie de Fourier

(1) Encontrar los coeficientes a_k , b_k de la serie de Fourier a través de la transformada de Fourier, comparar los resultados y complejidad de la resolución

1.

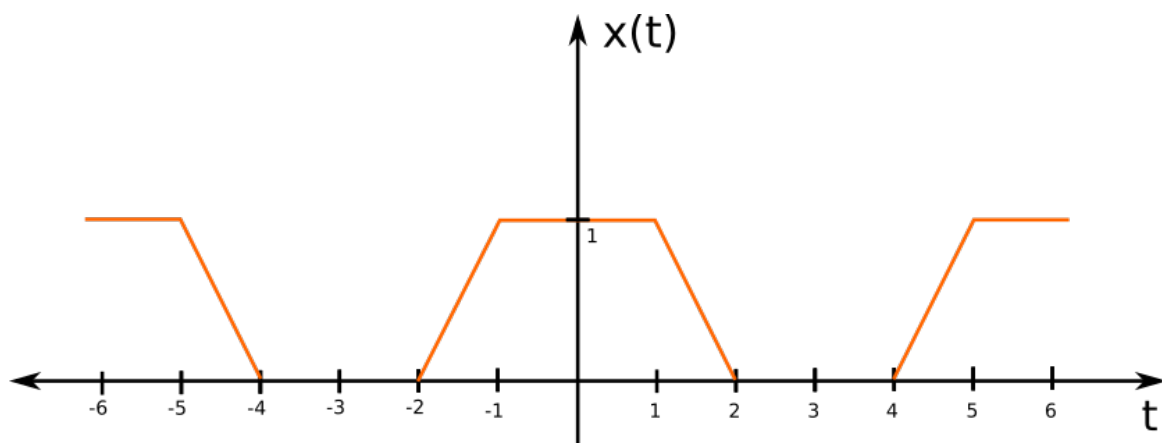


con $h(t) = e^{-4|t|}$

$$a_k = j \frac{2(-1)^k}{k\pi}, \quad a_0 = 0$$

$$b_k = \frac{16}{32 + (k\pi)^2} j \frac{2(-1)^k}{k\pi}$$

2.

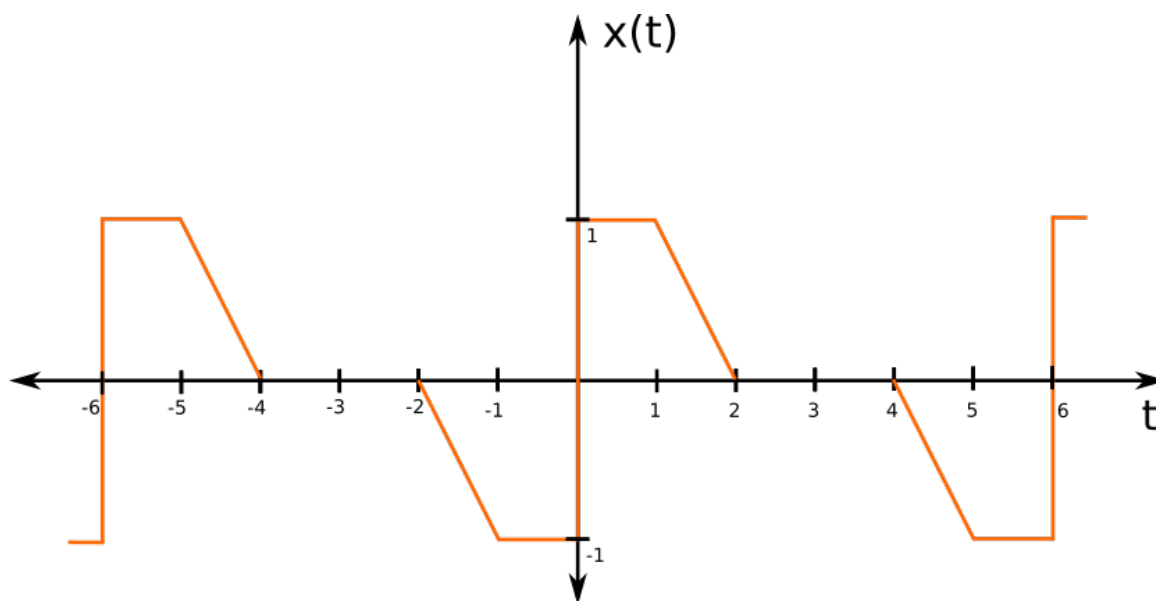


con $h(t) = \mu(t) \cdot e^{-2t}$

$$a_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2)$$

$$b_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) \frac{2}{4 + jk\pi}$$

3.

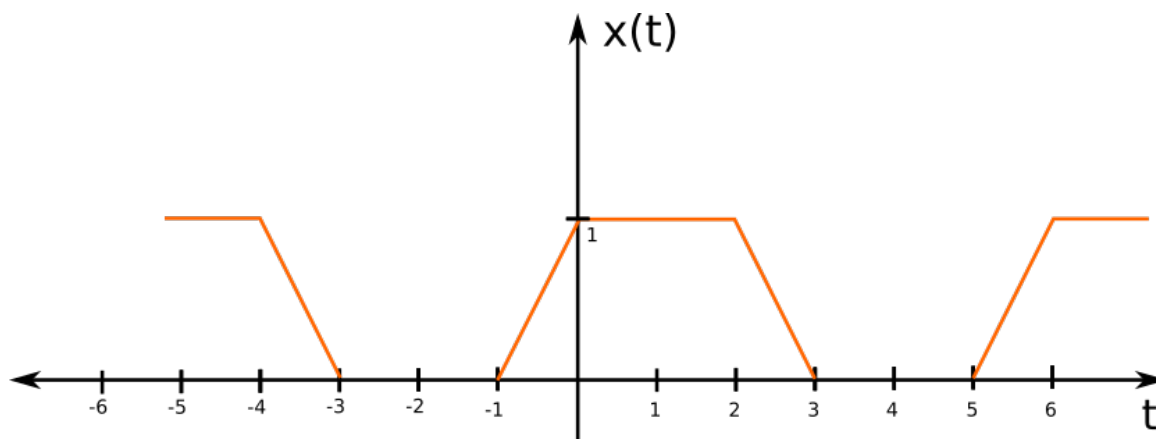


$$\text{con } h(t) = \delta(t - 3) + \delta(t + 3)$$

$$a_k = j \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \cos(k\pi/2)$$

$$b_k = 2(-1)^k a_k$$

4.



$$\text{con } h(t) = \mu(t) - \mu(t - 1)$$

$$a_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) e^{-jk\pi/3}$$

$$b_k = a_k \frac{8 \text{sen}(k\pi/4)}{k\pi} e^{-jk\pi/6} = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) e^{-jk\pi/3} \frac{8 \text{sen}(k\pi/4)}{k\pi} e^{-jk\pi/6}$$